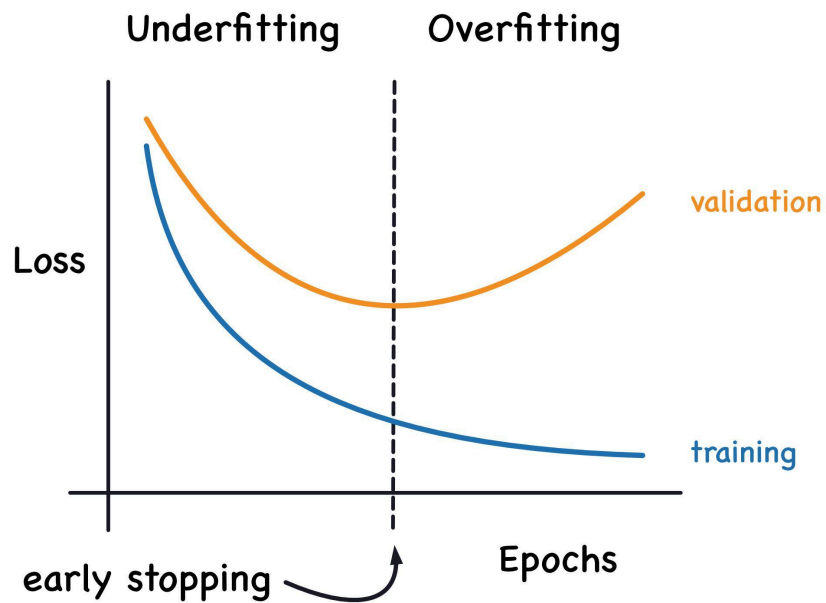
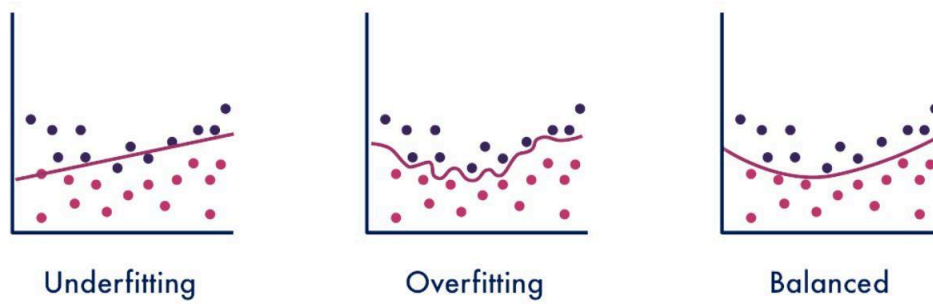
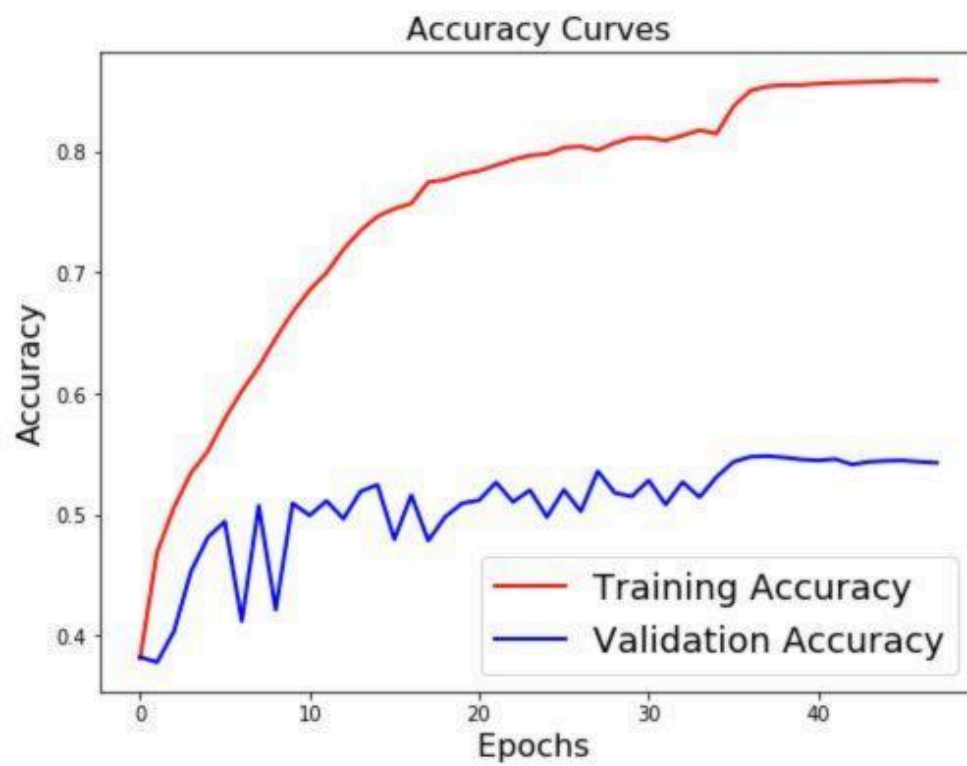
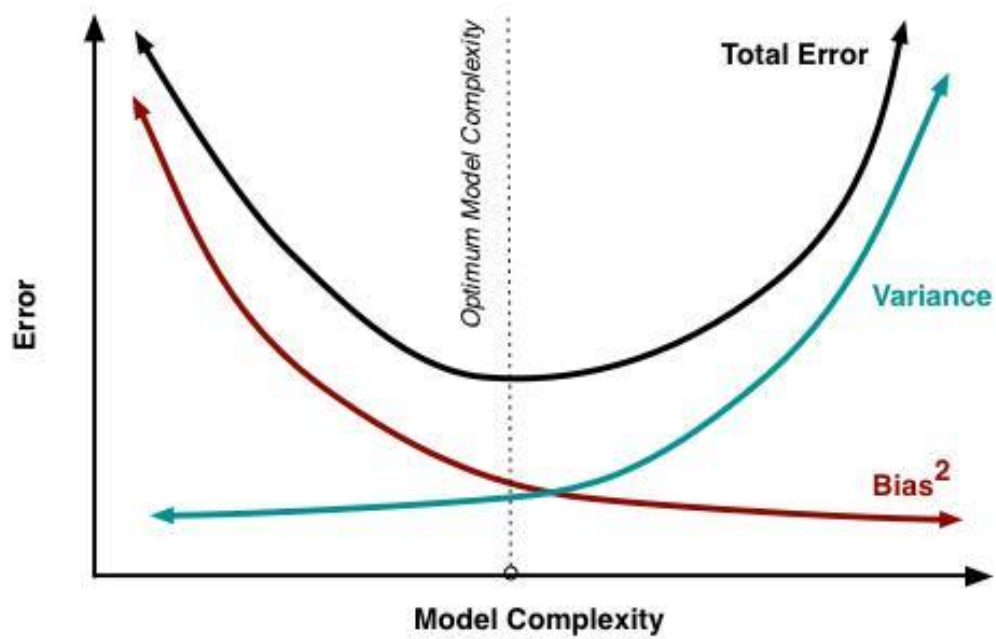


Reducing Overfitting With Data

What is Overfitting?





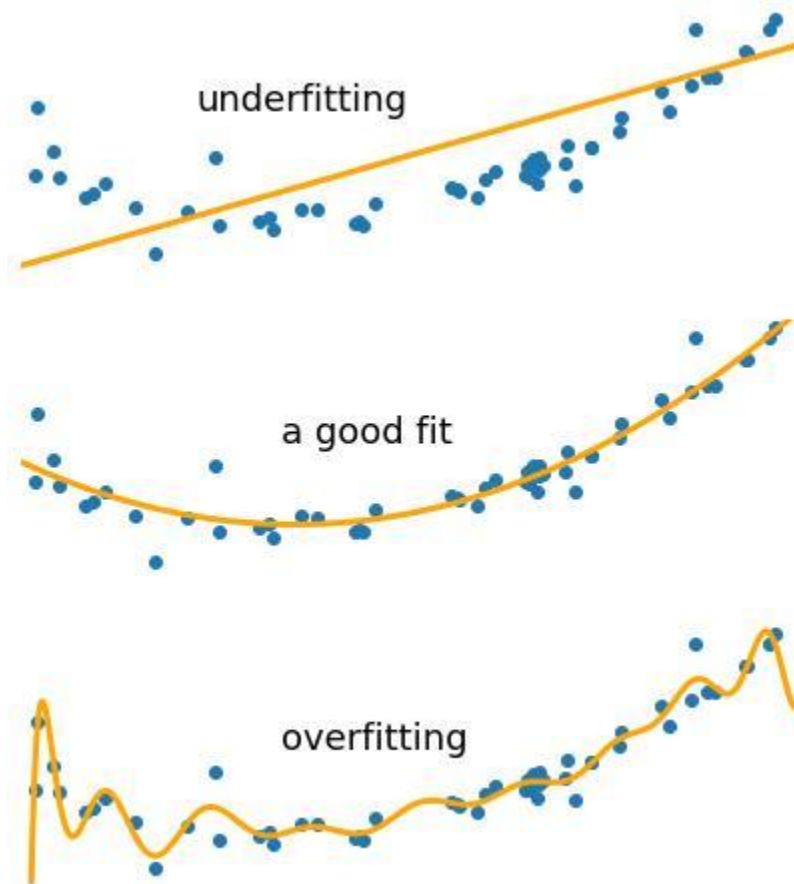
Overfitting happens when a machine learning model learns the training data **too well**, including noise and small details that are not important.

In simple words:

- The model performs **very well on training data**
- But performs **poorly on new (unseen) data**

It means the model **memorizes** instead of **learning patterns**.

Why Does Overfitting Happen?



Methods to Control Overfitting

Linear Model (Regression)	Tree-Based Algorithms (Decision Trees, Random Forest, XGBoost)	Neural Networks
Regularisation	Pruning	Early Stopping
Feature Selection	Ensembling	Dropout
Treating outliers	Regularisation	
	Early Stopping	

Overfitting usually occurs because of:

1. **Small Dataset** – Not enough examples to learn general patterns
2. **Too Complex Model** – Deep networks with too many parameters
3. **Noisy Data** – Random errors confuse the model
4. **Too Many Training Epochs** – Model keeps learning even after it should stop

How to Overcome Overfitting

There are many practical techniques to reduce overfitting. Below are the most important ones:

1. Collecting More Data

More data helps the model learn **general patterns instead of memorizing**.

Benefits:

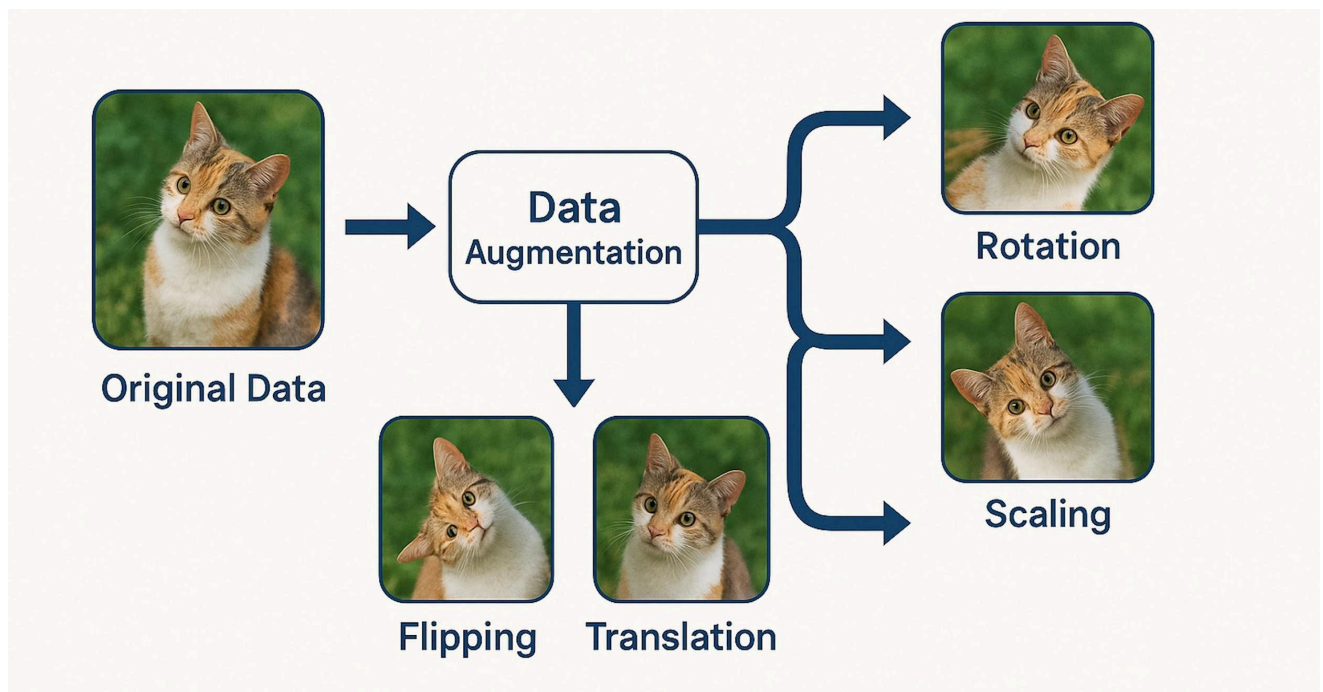
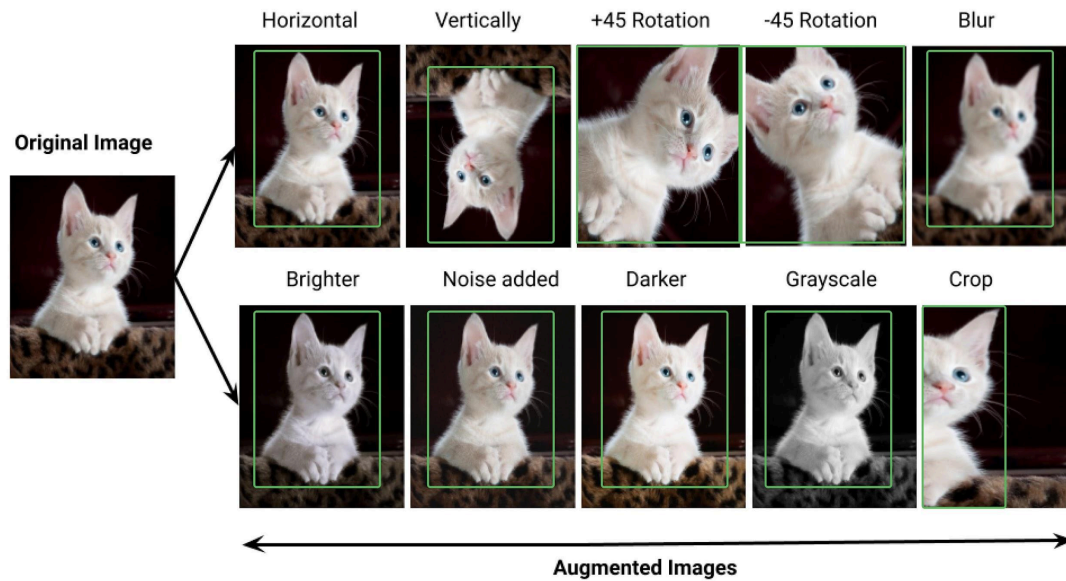
- Reduces variance
- Improves generalization
- Makes the model more robust

Example:

If you train a cat vs dog classifier with only 50 images, it will overfit.

But with **10,000+ images**, it learns better features.

2. Data Augmentation



Data augmentation artificially increases dataset size by modifying existing data.

Common techniques:

- Rotation
- Flipping
- Cropping
- Adding noise
- Color changes

Example (for images):

A single image can become **10+ different images** using augmentation.

This is very useful in **deep learning (CNN models)**.

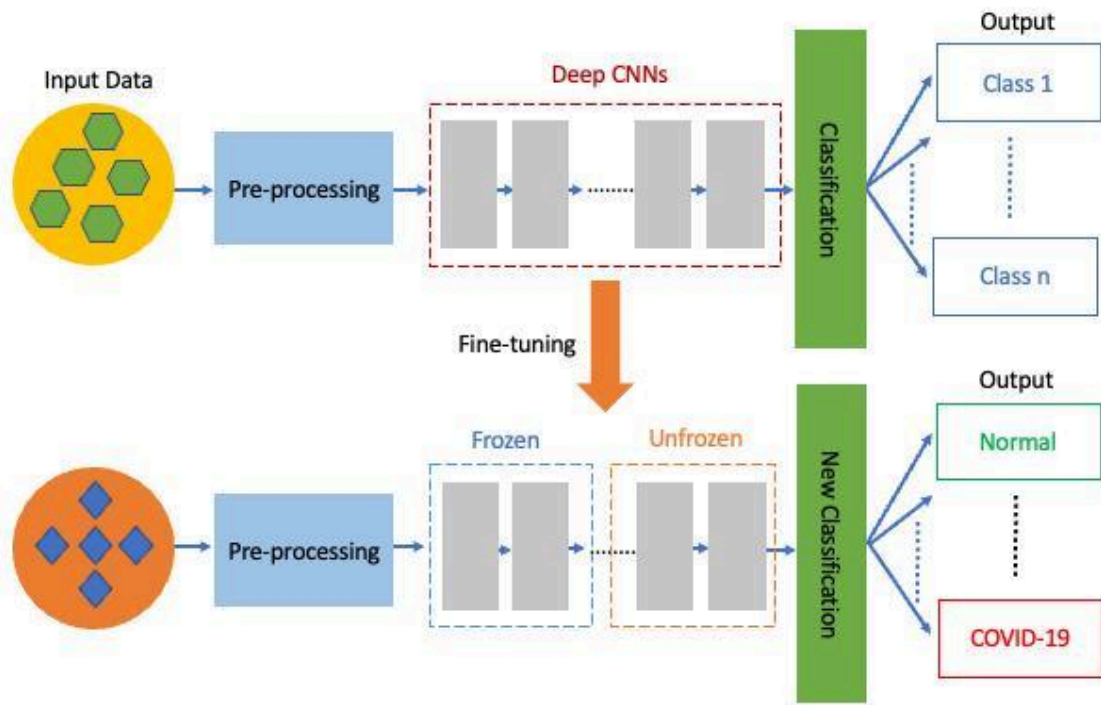
3. Pretraining (Transfer Learning)

Instead of training from scratch, we use a **pretrained model** (trained on large datasets like ImageNet).

Advantages:

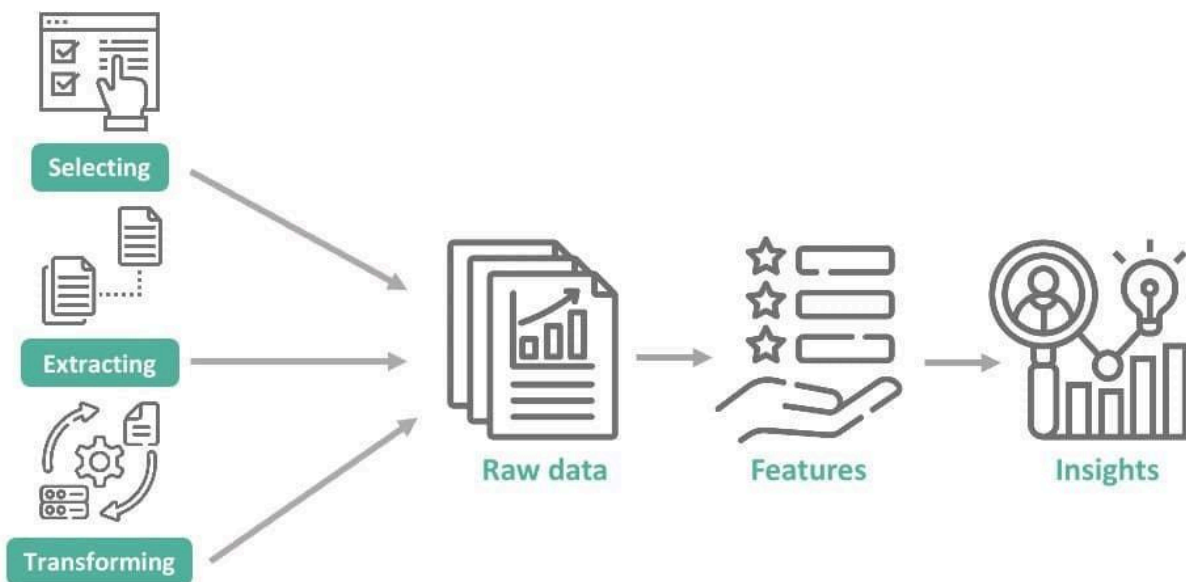
- Already learned general features
- Needs less data
- Reduces overfitting Example:
Use models like:
 - ResNet
 - EfficientNet
 - VGG

Then fine-tune them for your task.



4. Feature Engineering (FE)

What is Feature Engineering?



Feature Engineering means creating better input features for the model.

✓ Why important?

- Removes irrelevant data
- Keeps useful patterns
- Simplifies learning

Example:

- Convert date → day, month, year
- Normalize numerical values
- Remove noisy features

5. Normalization

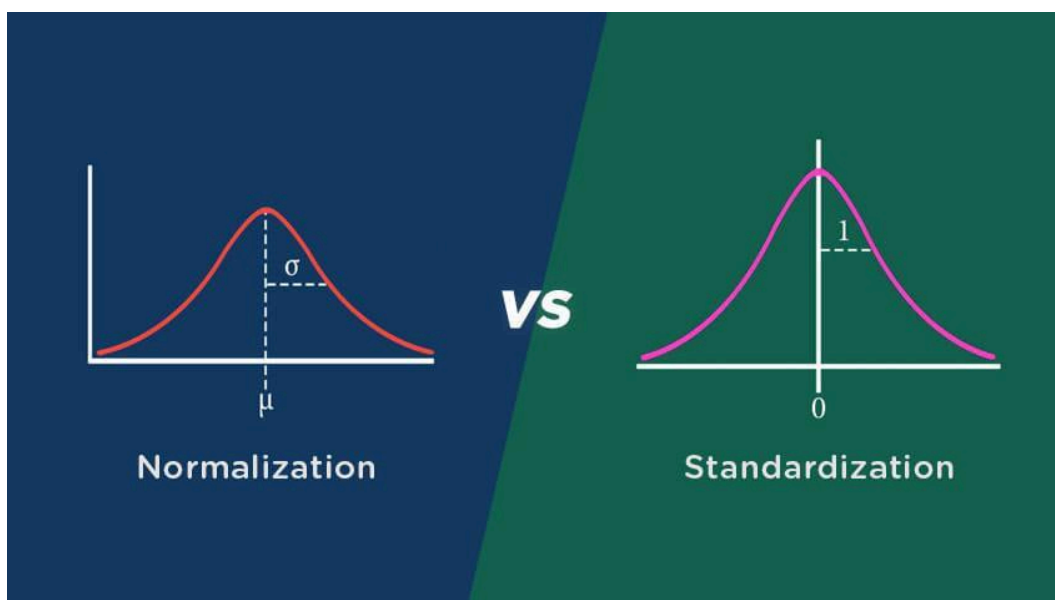
Normalization scales data into a fixed range (like 0 to 1).

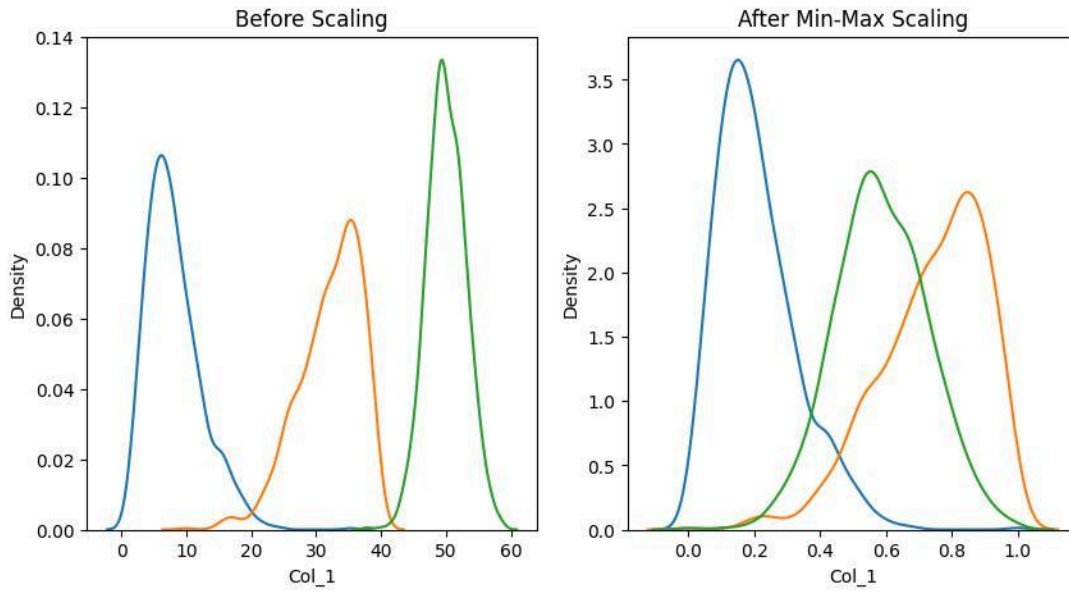
Benefits:

- Faster training
- Stable learning
- Prevents dominance of large values

Common methods:

- Min-Max Scaling
- Standardization (Z-score)





6. Smaller Batch Size

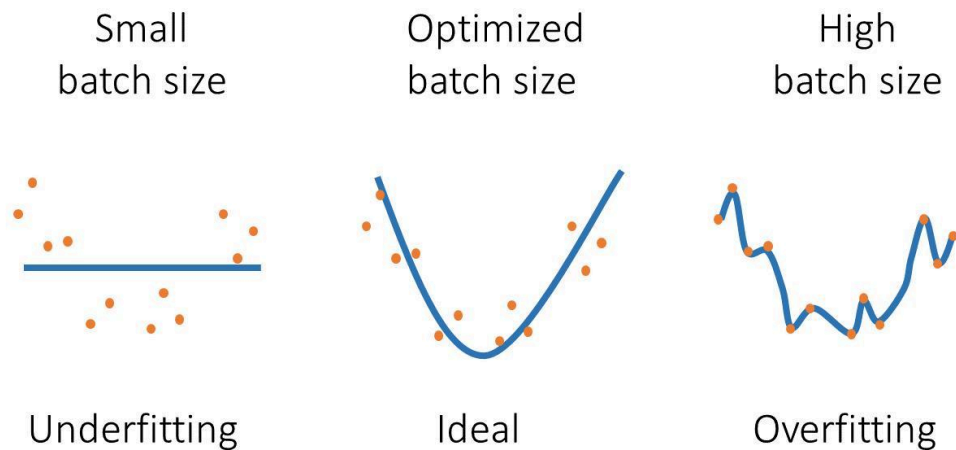
Batch size = number of samples used in one training step.

Smaller batch size helps because:

- Adds **noise in gradients**
- Prevents model from memorizing
- Improves generalization

Example:

- Batch size 32 → better generalization
- Batch size 1024 → may overfit



Small Batch v.s. Large Batch

	Small	Large
Speed for one update (no parallel)	Faster	Slower
Speed for one update (with parallel)	Same	Same (not too large)
Time for one epoch	Slower	Faster 
Gradient	Noisy	Stable
Optimization	Better 	Worse
Generalization	Better 	Worse

Batch size is a hyperparameter you have to decide.